Li et al. [81] added a L1 penalty to the optimization to encourage a sparse mapping between neurons. Bau et al. [10] matched the full representation of one model to a single neuron of another by linear regression.

Aligned Cosine Similarity. This measure was used to quantify similarity of instance representations, such as embeddings of individual words over time [54]. Its idea is to first align the representations by the orthogonal Procrustes transformation, and then to use cosine similarity (Eq. A.5) to measure similarity between the aligned representations. Letting $\boldsymbol{Q}^*$ denote the solution to the Procrustes problem (Eq. 15), the similarity of two instance representations is given by $\text{cos-sim}\left((\boldsymbol{R}\boldsymbol{Q}^*)_i, \boldsymbol{R}'_i\right)$. Overall similarity can then be analyzed by comparing the overall distribution of similarity scores, or aggregating them by, for instance, taking their mean value [124]. The latter option yields a similarity measure

$$m_{\text{Aligned-Cossim}}(\boldsymbol{R}, \boldsymbol{R}') = \frac{1}{N}\sum_{i=1}^{N} \text{cos-sim}\left((\boldsymbol{R}\boldsymbol{Q}^*)_i, \boldsymbol{R}'_i\right), \tag{20}$$

which is bounded in the interval $[-1, 1]$, with $m_{\text{Aligned-Cossim}}(\boldsymbol{R}, \boldsymbol{R}') = 1$ indicating perfect similarity. It is invariant to orthogonal transformations and isotropic scaling.

Correlation Match. Li et al. [81] measured representational similarity by creating a correlation matrix between the neuron activations of two representations that are assumed to be mean-centered. They then matched each neuron $\boldsymbol{R}_{-,j}$ to the neuron $\boldsymbol{R}'_{-,k}$ that it correlated the strongest with. Wu et al. [156] applied strict one-to-one matching, Li et al. [81] further used a relaxed version, in which one neuron can correspond to multiple other ones. Letting $\boldsymbol{M}$ denote the matrix that matches the neurons, which is a permutation matrix in strict one-to-one matching, the average correlation between the matched neurons is given by

$$m_{\text{Corr-Match}}(\boldsymbol{R}, \boldsymbol{R}') = \frac{1}{D}\sum_{j=1}^{D} \frac{\langle \boldsymbol{R}_{-,j}, (\boldsymbol{R}'\boldsymbol{M})_{-,j}\rangle}{\|\boldsymbol{R}_{-,j}\|_2 \|(\boldsymbol{R}'\boldsymbol{M})_{-,j}\|_2}, \tag{21}$$

This measure is invariant to permutations, isotropic scaling, and translations. A value of one indicates equivalent representations, a value of zero uncorrelated ones.

Maximum Matching Similarity. In contrast to the previous measures, *Maximum Matching Similarity* [152] aligns representations only implicitly and can compare representations of different dimension by testing whether neuron activations of one representation, i.e., columns of the representation matrix, (approximately) lie in a subspace spanned from neuron activations of the other representation. Every neuron, of which the activation vector can be approximated by such a subspace, is then considered part of a match between the representations. Following this intuition, the main idea of the measure proposed by Wang et al. [152] is to find the maximal set of neurons in each representation that can be matched with the other subspace. Formally, for an index subset $\mathcal{J} \subseteq \{1, \ldots, D\}$, let $\boldsymbol{R}_{-,\mathcal{J}} = \{\boldsymbol{R}_{-,j}, j \in \mathcal{J}\}$ denote the set of corresponding neuron activation vectors. Then a pair $(\mathcal{J}, \mathcal{J}')$ forms an ε-approximate match, $\varepsilon \in (0, 1]$, on the representations $\boldsymbol{R}, \boldsymbol{R}'$ if for all $j \in \mathcal{J}, j' \in \mathcal{J}'$ it holds that

$$\min_{\boldsymbol{r} \in \text{span}(\boldsymbol{R}_{-,\mathcal{J}})} \|\boldsymbol{R}'_{-,j'} - \boldsymbol{r}\| \leq \varepsilon \cdot \|\boldsymbol{R}'_{-,j'}\| \quad \text{and} \quad \min_{\boldsymbol{r}' \in \text{span}(\boldsymbol{R}'_{-,\mathcal{J}'})} \|\boldsymbol{R}_{-,j} - \boldsymbol{r}'\| \leq \varepsilon \cdot \|\boldsymbol{R}_{-,j}\|. \tag{22}$$

A pair $(\mathcal{J}_{\max}, \mathcal{J}'_{\max})$ is considered a maximum match, if for all ε-matches $(\mathcal{J}, \mathcal{J}')$ it holds that $\mathcal{J} \subseteq \mathcal{J}_{\max}$ and $\mathcal{J}' \subseteq \mathcal{J}'_{\max}$. Wang et al. [152] showed that the maximum match is unique and provided algorithms to determine it. Based on the maximum match, the *maximum matching similarity* is defined as

$$m^{\varepsilon}_{\text{maximum-match}}(\boldsymbol{R}, \boldsymbol{R}') = \frac{|\mathcal{J}_{\max}| + |\mathcal{J}'_{\max}|}{D + D'}. \tag{23}$$

This measure is invariant to invertible linear transformation, since such transformations do not alter the subspaces. It is bounded in the interval $[0, 1]$, with a similarity score of 1 indicating maximum similarity.

ContraSim. Inspired by ideas from contrastive learning, Rahamim and Belinkov [116] proposed a measure that implicitly aligns representations by applying a neural encoder to map them into a joint embedding space. The encoder was trained using explicitly selected pairs of instances as positive and negative samples, for which the resulting embeddings should and should not be similar, respectively. In the joint embedding space, similarity is then modeled by the angle between representations, and thus, letting enc denote the trained encoder network, *ContraSim* is defined as

$$m_{\text{ContraSim}}(\boldsymbol{R}, \boldsymbol{R}') = \frac{1}{N}\sum_{i=1}^{N} \text{cos-sim}\left(\text{enc}(\boldsymbol{R}_i), \text{enc}(\boldsymbol{R}'_i)\right). \tag{24}$$

If $\boldsymbol{R}, \boldsymbol{R}'$ have different dimensionality, two different encoders are trained together. The invariances of the measure are determined via the training examples for the encoder. The measure is bounded in the interval $[-1, 1]$, with a similarity score of 1 indicating maximum similarity.

3.3 Representational Similarity Matrix-Based Measures

A common approach to avoid alignment issues in direct comparisons of representations is to use *representational similarity matrices* (RSMs). Intuitively, an RSM describes the similarity of the representation of each instance i to

all other instances in a given representation $\boldsymbol{R}$. The RSMs of two representations $\boldsymbol{R}, \boldsymbol{R}'$ can then be used to quantify representational similarity in terms of the difference between these RSMs. Formally, given an instance-wise similarity function $s : \mathbb{R}^D \times \mathbb{R}^D \longrightarrow \mathbb{R}$, the RSM $\boldsymbol{S} \in \mathbb{R}^{N \times N}$ of a representation $\boldsymbol{R}$ can be defined in terms of its elements via

$$\boldsymbol{S}_{i,j} := s(\boldsymbol{R}_i, \boldsymbol{R}_j). \tag{25}$$

Each row $\boldsymbol{S}_i$ then corresponds to the similarity between the representations of instance i and the representations of all other inputs, including itself. RSMs can be computed with a variety of similarity functions s such as cosine similarity [24] or kernel functions [73]—like before, we do not differentiate between the equivalent concepts of similarity and distance functions. Naturally, the choice of the underlying similarity function s impacts the kind of transformations that the representational similarity measures m will be invariant to: if the RSM is unchanged by a transformation, then the representational similarity will not change either. In Appendix A.3 we give an overview of commonly used similarity functions, along with the invariances they induce on the RSMs. After selecting a suitable similarity function s, two RSMs $\boldsymbol{S}, \boldsymbol{S}'$ are compared. In the following, we review existing measures that use this approach.

Norm of RSM Difference. A direct approach to compare RSMs is to apply some matrix norm $\| \cdot \|$ to the difference between RSMs to obtain a measure

$$m_{\text{Norm}}(\boldsymbol{R}, \boldsymbol{R}') = \|\boldsymbol{S} - \boldsymbol{S}'\|. \tag{26}$$

which assigns a score of zero to equivalent representations, and higher scores to dissimilar representations. To compute the RSMs, Shahbazi et al. [125] and Yin and Shen [162] used the linear kernel. In that case, this measure is invariant to orthogonal transformations and satisfies the properties of a distance metric for representations of equal dimensionality.

Representational Similarity Analysis. Kriegeskorte et al. [74] proposed *Representational Similarity Analysis* (RSA) in neuroscience. RSA is a general framework that utilizes RSMs to compare sets of measurements, such as neural representations. In the first step of this framework, RSMs are computed with respect to an inner similarity function s_{in}. Since the RSMs are symmetric, their lower triangles can then be vectorized in a next step to vectors $\mathsf{v}(\boldsymbol{S}) \in \mathbb{R}^{N(N-1)/2}$. Finally, these vectors are compared by an outer similarity function s_{out}:

$$m_{\text{RSA}}(\boldsymbol{R}, \boldsymbol{R}') = s_{\text{out}}(\mathsf{v}(\boldsymbol{S}), \mathsf{v}(\boldsymbol{S}')). \tag{27}$$

This framework can be instantiated with various choices for the similarity functions s_{in} and s_{out}. This choice however affects the kind of transformations that RSA is invariant to, and further determines the range and interpretation of this measure. Kriegeskorte et al. [74] used Pearson correlation (Eq. A.8) as inner similarity function s_{in} to compute the RSMs, and Spearman correlation as outer similarity function s_{out}, since these correlation measures induce invariance to scaling and translations. Kriegeskorte et al. [74] further suggested functions such as Euclidean or Mahalanobis distance.

Centered Kernel Alignment. Kornblith et al. [73] proposed *Centered Kernel Alignment* (CKA) [28, 29] to measure representational similarity. CKA uses kernel functions on mean-centered representations to compute the RSMs, which are then compared via the Hilbert-Schmidt Independence Criterion (HSIC) [50]. Given two RSMs $\boldsymbol{S}, \boldsymbol{S}'$, the HSIC can be computed via $\text{HSIC}(\boldsymbol{S}, \boldsymbol{S}') = \frac{1}{(N-1)^2} \operatorname{tr}(\boldsymbol{S}\boldsymbol{H}_N\boldsymbol{S}'\boldsymbol{H}_N)$, where $\boldsymbol{H}_N = \boldsymbol{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\mathsf{T}$ denotes a centering matrix. Recent work [103] highlights the importance of using the debiased HSIC estimator of Song et al. [132], especially when $N < D$. Then, a normalization of the HSIC yields the CKA measure:

$$m_{\text{CKA}}(\boldsymbol{R}, \boldsymbol{R}') = \frac{\text{HSIC}(\boldsymbol{S}, \boldsymbol{S}')}{\sqrt{\text{HSIC}(\boldsymbol{S}, \boldsymbol{S})\text{HSIC}(\boldsymbol{S}', \boldsymbol{S}')}}. \tag{28}$$

CKA is bounded in the interval $[0, 1]$, with $m_{\text{CKA}}(\boldsymbol{R}, \boldsymbol{R}') = 1$ indicating equivalent representations. Kornblith et al. [73] computed the RSMs from the linear kernel and tested the RBF kernel without reporting large differences in results. Saini et al. [122] used so-called affinity matrices, which result from sparse subspace clustering [38] of the representations, instead of RSMs. The standard linear version is invariant to orthogonal transformations and isotropic scaling.

CKA with linear kernel is equivalent to the RV coefficient, a statistical measure to compare data matrices [120, 73]. It can also be seen as a variant of PWCCA (Eq. 14) with an alternative weighting scheme, with the advantage that it does not require a matrix decomposition to be computed [73]. Further, Godfrey et al. [47] proposed the G_{ReLU}-CKA variant that is specific to models that use ReLU activations, and invariant to G_{ReLU} transformations (Eq. A.10).

Distance Correlation. *Distance Correlation* (dCor) [139] is a non-linear correlation measure that tests dependence of two vector-valued random variables X and Y with finite mean. In the context of our survey, we consider the instance representations as samples of such random variables. To determine the distance correlation of two representation matrices $\boldsymbol{R}, \boldsymbol{R}'$, one first computes the RSMs $\boldsymbol{S}, \boldsymbol{S}'$ using Euclidean distance as similarity function s. Next, the RSMs are mean-centered in both rows and columns, which yields $\widetilde{\boldsymbol{S}}, \widetilde{\boldsymbol{S}}'$. Then the squared sample distance covariance of the RSMs $\boldsymbol{S}, \boldsymbol{S}'$ can be computed via $\text{dCov}^2(\boldsymbol{S}, \boldsymbol{S}') = \frac{1}{N^2}\sum_{i=1}^N \sum_{j=1}^N \widetilde{\boldsymbol{S}}_{i,j}\widetilde{\boldsymbol{S}}'_{i,j}$. Finally, the squared distance correlation is defined as

$$m^2_{\text{dCor}}(\boldsymbol{R}, \boldsymbol{R}') = \frac{\text{dCov}^2(\boldsymbol{S}, \boldsymbol{S}')}{\sqrt{\text{dCov}^2(\boldsymbol{S}, \boldsymbol{S})\,\text{dCov}^2(\boldsymbol{S}', \boldsymbol{S}')}}. \tag{29}$$

A distance correlation of zero indicates statistical independence between the representations $\boldsymbol{R}$ and $\boldsymbol{R'}$. Due to the usage of Euclidean distance as similarity function s, dCor is invariant to orthogonal transformations and translations.

Lin [83] considered a variant called *Adaptive Geo-Topological Independence Criterion* (AGTIC) [84], which rescales the values in the RSMs with respect to an upper and lower threshold to eliminate noise.

Normalized Bures Similarity. This measure was inspired by the Bures distance, which has its roots in quantum information theory [20] and satisfies the properties of a distance metric on the space of positive semi-definite matrices [13]. As Tang et al. [141] used the linear kernel to compute the RSMs $\boldsymbol{S}, \boldsymbol{S'}$, these matrices are positive semi-definite. Hence, these matrices also have a unique square root. Therefore, they could define the *Normalized Bures Similarity* as

$$m_{\text{NBS}}(\boldsymbol{R}, \boldsymbol{R'}) = \frac{\operatorname{tr}(\boldsymbol{S}^{1/2}\boldsymbol{S'}\boldsymbol{S}^{1/2})^{1/2}}{\sqrt{\operatorname{tr}(\boldsymbol{S})\operatorname{tr}(\boldsymbol{S'})}}. \tag{30}$$

This measure is bounded in the interval $[0, 1]$, with $m_{\text{NBS}}(\boldsymbol{R}, \boldsymbol{R'}) = 1$ indicating perfect similarity. Due to use of the linear kernel, it is invariant to orthogonal transformations, and further invariant to isotropic scaling due to the normalization.

One can show that NBS is equivalent—up to an arc cosine—to the angular shape metric (Eq. 16), and that the unnormalized Bures distance is equal to the orthogonal Procrustes measure [55].

Eigenspace Overlap Score. May et al. [95] proposed the *Eigenspace Overlap Score* (EOS) as a criterion to select compressed word embeddings with best downstream performance. EOS compares RSMs by comparing the spaces spanned from their eigenvectors. Assuming full-rank representations $\boldsymbol{R}, \boldsymbol{R'}$, they compute the RSMs $\boldsymbol{S}, \boldsymbol{S'}$ using the linear kernel (Eq. A.6). Letting $\boldsymbol{U} \in \mathbb{R}^{N \times D}, \boldsymbol{U'} \in \mathbb{R}^{N \times D'}$ denote the matrices of eigenvectors that correspond to the non-zero eigenvalues of $\boldsymbol{S}, \boldsymbol{S'}$, respectively, the measure is defined as

$$m_{\text{EOS}}(\boldsymbol{R}, \boldsymbol{R'}) = \frac{1}{\max(D, D')} \|\boldsymbol{U}^\top \boldsymbol{U'}\|_F^2. \tag{31}$$

EOS indicates minimal similarity with a value of zero when the spans of $\boldsymbol{U}$ and $\boldsymbol{U'}$ are orthogonal, and maximal similarity with a value of one when the spans are identical. This measure is invariant to invertible linear transformations.

EOS is related to the expected difference in generalization error of two linear models that are each trained on one of the representations [95], similar to the following measure.

Unified Linear Probing (GULP). *GULP* quantifies similarity by measuring how differently linear regression models that use either the representation $\boldsymbol{R}$ or the representation $\boldsymbol{R'}$ [17] can generalize. This is done by considering all regression functions η on the original instances $\boldsymbol{X}$ that are bounded so that $\|\eta\|_{L^2} \leq 1$, and trying to replicate these relations via ridge regression on the representations $\boldsymbol{R}$ and $\boldsymbol{R'}$. The similarity measure is then defined as the supremum of the expected discrepancy in the predictions of ridge regression models trained to approximate η with $\boldsymbol{R}$ or $\boldsymbol{R'}$ as inputs, taken over all regression functions η.

Practically, Boix-Adsera et al. [17] proved that there is a closed form expression to estimate this value in terms of the covariance matrices of the representations, where it is assumed that the representations are mean-centered in the columns, and that their rows have unit norm. Letting the RSMs $\boldsymbol{S} = \frac{1}{N}\boldsymbol{R}^\top\boldsymbol{R}$ denote the matrix of covariance within a representation, $\boldsymbol{S}_{\boldsymbol{R},\boldsymbol{R'}} = \frac{1}{N}\boldsymbol{R}^\top\boldsymbol{R'}$ the cross-covariance matrix, and $\boldsymbol{S}^{-\lambda} = (\boldsymbol{S} + \lambda\boldsymbol{I}_D)^{-1}$ the inverse of a regularized covariance matrix, the GULP measure can be computed as

$$m^{\lambda}_{\text{GULP}}(\boldsymbol{R}, \boldsymbol{R'}) = \Big(\operatorname{tr}(\boldsymbol{S}^{-\lambda}\boldsymbol{S}\boldsymbol{S}^{-\lambda}\boldsymbol{S}) + \operatorname{tr}(\boldsymbol{S'}^{-\lambda}\boldsymbol{S'}\boldsymbol{S'}^{-\lambda}\boldsymbol{S'}) - 2\operatorname{tr}(\boldsymbol{S}^{-\lambda}\boldsymbol{S}_{\boldsymbol{R},\boldsymbol{R'}}\boldsymbol{S'}^{-\lambda}\boldsymbol{S}_{\boldsymbol{R},\boldsymbol{R'}}^\top)\Big)^{1/2}. \tag{32}$$

The hyperparameter $\lambda \geq 0$ corresponds to the regularization weight of the ridge regression models over the representations. For all $\lambda \geq 0$, GULP is unbounded, satisfies the properties of a distance metric, and is invariant to orthogonal transformations, scaling, and translations. For $\lambda = 0$, GULP is invariant to affine transformations, and can further be expressed as a linear transformation of the mean-squared CCA measure (Eq. 12).

Transferred Discrepancy (TD) [41] used an approach similar to GULP by measuring the discrepancy of linear classifiers, instead of linear regression models. TD is also related to the mean-squared CCA measure (Eq. 12).

Riemannian Distance. This measure considers the special geometry of symmetric positive definite (SPD) matrices, which lie on a Riemannian manifold [12]. Every inner product defined on a Riemannian manifold induces a distance metric that considers the special curvature of these structures. On the manifold of SPD matrices,

$$m_{\text{Riemann}}(\boldsymbol{R}, \boldsymbol{R'}) = \sqrt{\textstyle\sum_{i=1}^{N} \log^2(\lambda_i)}, \tag{33}$$

denotes such a metric, where λ_i is the i-th eigenvalue of $\boldsymbol{S}^{-1}\boldsymbol{S'}$. Shahbazi et al. [125] proposed this measure using RSMs defined as $\boldsymbol{S} = \boldsymbol{R}\boldsymbol{R}^\top/D$. This matrix however can only be positive definite if $D > N$, which limits applicability